

Negative Exponent Law

Teacher's Edition

The Period

MA.8.NSO.1.3 Extend previous understanding of the laws of exponents to include integer exponents. Apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational-number bases, with procedural fluency.

Must Integer exponents, negatives included. Apply the laws to evaluate numerical expressions and to generate equivalent ones, with procedural fluency.

Must not No fractional exponents. Bases are nonzero wherever a negative exponent appears; rational bases wait for 3.05.

Target I can rewrite a power with a negative exponent using a positive exponent, and evaluate it.

MTR **MTR.5.1** — Notes I · **MTR.4.1** — Board 5 · **MTR.6.1** — Board 1

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2-3	5 min
Notes	4-6	11 min
Example 1	7-8	3 min
Your Turn	9-10	3 min
Example 2	11-12	3 min
Your Turn	13-14	3 min
Whiteboards	15-32	13 min
Independent Set	33	6 min
IXL	34	5 min
Total		53 min

Say these three things out loud today

1. A negative exponent asks for a reciprocal. It never asks for a negative number.
2. The base keeps its own sign. The exponent only tells you to flip.
3. Every law from last week still works — the exponents are just integers now.

Slide by Slide

1. **Negative Exponent Law.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q4 is the whole lesson in one question: a negative exponent asks for the RECIPROCAL of a power, and the misconception is that it asks for the opposite.
3. **Warm-Up — Answers.** Reveal. 1. 25 2. 5.5 3. $x = -7$ 4. $\frac{4}{3}$ and $-\frac{3}{4}$
4. **Notes — Copy the table.** Negative exponent, positive value. The sign of the exponent tells you to divide; it says nothing about the sign of the answer.
5. **Notes.** Reciprocal, not opposite. 5^{-2} is one twenty-fifth. It is a small positive number, not a negative one.
6. **Notes.** Two roads again: apply the negative exponent law first and multiply the reciprocal out, or apply the product law directly by adding a negative.
7. **Example 1 — Negative exponents top and bottom.** SPOKEN ONLY.
8. **Example 1 — Worked.** The 2^{-3} moves down and becomes 2^3 ; the 9^{-2} moves up and becomes 9^2 .
9. **Your Turn — Same steps, your numbers. Boards up when done.** Catches the sign-dropper (9/5) and the sign-flipper ($-5/9$).
10. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
11. **Example 2 — A quotient law crossing zero.** SPOKEN ONLY.
12. **Example 2 — Worked both ways.** Road 1: subtract the exponents, $0 - (-4) = 4$.
13. **Your Turn — Same steps, your numbers. Boards up when done.** Catches $-4 - (-6)$ done as -10 (a board with $1/(-8)^{10}$) and a dropped sign on the base (a board with $-64 -$ but $(-8)^2$ is 64).
14. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
- 15-16. **Board 1.** $8^{-3} \rightarrow \frac{1}{8^3} -8^3$ or -512 — *negative exponent read as a negative value [B1G-M 8.NSO.1.3 task: $-b$ vs b^{-1}]; $1/8$ — the exponent was dropped*
- 17-18. **Board 2.** A sheet of paper 30 centimeters long is folded in half 7 times. Each fold makes the top layer half as big. What fraction of the whole sheet is the top layer after 7 folds? $\rightarrow \frac{1}{128}$ *The length of 30 centimeters is not needed. 128 — the number of layers, not the fraction asked for [the question names the thing wanted]; $1/14$ — multiplied base by exponent; -128 — negative exponent read as a negative value [B1G-M 8.NSO.1.3]; anything built from 30 — the unneeded figure was used*
- 19-20. **Board 3.** $\frac{1}{10^{-2}} \rightarrow 10^2 = 100$ $1/100$ or 0.01 — *the negative exponent was made positive without moving the power [law applied in only one direction]*
- 21-22. **Board 4.** $(-2)^{-3} \rightarrow -\frac{1}{8}$ $1/8$ — *the base's own sign was dropped [8.NSO.1.3: rational bases include negatives]; -8 — the negative exponent was treated as positive*
- 23-24. **Board 5.** 5^{-2} Which is the value? $\rightarrow B - 1/25$ *(A) reciprocal of $5 \cdot 2$ instead of 5^2 [reciprocal right, power wrong]; (C) the negative exponent was read as a negative value, $-(5^2)$ [B1G-M 8.NSO.1.3 task: $-b$ vs b^{-1}]; (D) multiplied the base by the*

exponent, $5 \cdot (-2)$ [misconception: exponent treated as a factor] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.

25-26. Board 6. $6^{-2} \cdot 6^5$ Which expression is equivalent? → **D — 6^3** (A) multiplied the exponents [confused with power of a power]; (B) multiplied the bases as well [7.NSO.1.1 misconception: operating on the bases]; (C) subtracted instead of adding, $-2 - 5$ [product law: add the exponents, including a negative one] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.

27-28. Board 7. $3^{-4} \cdot 3^4$ → **1** 0 — zero exponent read as 0 [8.NSO.1.3: zero exponent law, $a^0 = 1$]; 81 or 1/81 — one factor was ignored

29-30. Board 8. $\frac{9^{-5}}{9^3}$ → $\frac{1}{9^8}$ 9^{-2} or 1/81 — subtracted 5 — 3 ignoring the sign [integer subtraction, MA.7 prerequisite]; 9^8 — the minus was dropped at the end

31-32. Board 9. Ellen says the value of 5^{-2} is -25 . Explain why she is wrong, and show the correct value. → **$5^{-2} = 1/5^2 = 1/25$** ' $-1/25$ ' — took the reciprocal AND the opposite [B1G-M 8.NSO.1.3 task: $-b$ vs b^{-1}]; 'she forgot parentheses' — confused 5^{-2} with $(-5)^2$

33. Independent Set — Six questions. On your own, in silence. Hand out the Independent Set.

34. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Misconceptions to Watch

Board 1. -8^3 or -512 — negative exponent read as a negative value [B1G-M 8.NSO.1.3 task: $-b$ vs b^{-1}]

Board 2. 128 — the number of layers, not the fraction asked for [the question names the thing wanted]

Board 3. 1/100 or 0.01 — the negative exponent was made positive without moving the power [law applied in only one direction]

Board 4. 1/8 — the base's own sign was dropped [8.NSO.1.3: rational bases include negatives]

Board 5. reciprocal of $5 \cdot 2$ instead of 5^2 [reciprocal right, power wrong]

Board 6. multiplied the exponents [confused with power of a power]

Board 7. 0 — zero exponent read as 0 [8.NSO.1.3: zero exponent law, $a^0 = 1$]

Board 8. 9^{-2} or 1/81 — subtracted 5 — 3 ignoring the sign [integer subtraction, MA.7 prerequisite]

Board 9. ' $-1/25$ ' — took the reciprocal AND the opposite [B1G-M 8.NSO.1.3 task: $-b$ vs b^{-1}]

Materials Whiteboards and markers. Calculators are allowed on everything; on a calculator $5^{(-2)}$ returns 0.04, which is 1/25 and worth showing once.